

Adoption of Compliant Authoring Tools in Doctoral Education: Three Studies

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Chapter 1

Preface

This page exists only to hold project metadata for Quarto.

Chapter 2

List of Publications

The work in this thesis has produced the following publications and manuscripts. The candidate is the first author of all three papers; co-author contributions are documented in each constituent chapter.

Note

Replace the placeholders below with your own publication list. The University of Manchester policy on Journal Format theses (section 13) encourages candidates to indicate, for each paper, its publication status (published / submitted / in preparation) and the candidate's contribution.

1. **Reader, Avery & Co-author, A.** (2026). *Adoption patterns of compliant authoring tools in UK business schools*. Journal of Doctoral Education. (Chapter 2.)
2. **Reader, Avery, Co-author, A. & Co-author, B.** (2026). *Compliance enforcement and writing quality: An A/B experiment*. Manuscript under review. (Chapter 3.)
3. **Reader, Avery & Co-author, C.** (2026). *Supervisor perspectives on tooling in doctoral supervision*. Manuscript in preparation. (Chapter 4.)

Chapter 3

Abstract

This thesis examines the adoption of policy-compliant authoring tools in doctoral education through three constituent papers, submitted in **journal format** per Presentation of Theses Policy section 13. The papers address distinct but complementary aspects of the central question: a cross-institutional survey of tool adoption (Paper 1), an experimental study of compliance enforcement on writing quality (Paper 2), and a qualitative analysis of supervisor perspectives (Paper 3).

Together, the papers establish that policy-compliant tooling is associated with measurable improvements in submission quality, that this association is mediated by supervisor engagement, and that adoption patterns vary systematically with disciplinary culture. The conclusion synthesises across the three papers and offers practical recommendations.

All findings are reproducible from the underlying data and code, made available alongside the submitted PDF.

Chapter 4

Declaration of Originality

I declare that no portion of the work referred to in this thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Chapter 5

Copyright Statement

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Avery L. Reader

Chapter 6

Acknowledgements

I would like to thank my supervisors for their guidance, my examiners for their constructive feedback, and my co-authors for their collaboration on the constituent papers presented in this thesis.

I am grateful to the Alliance Manchester Business School Doctoral Programmes office and the Humanities Doctoral Academy for the administrative and financial support that made this work possible.

Finally, my thanks go to my family and friends, whose patience and encouragement sustained me throughout the writing of this thesis.

Part I

Main body

Chapter 7

Introduction

This thesis is submitted in **journal format** under the University of Manchester *Presentation of Theses Policy* section 13. The format permits doctoral candidates to organise the body of the thesis around a series of self-contained papers, framed by an overarching introduction and conclusion.

7.1 Rationale for journal format

The decision to submit in journal format reflects three considerations:

1. **Disciplinary norms.** Research in management and organisational behaviour is typically disseminated through peer-reviewed journal articles rather than monographs. Submitting in journal format aligns the thesis with the conventions of the field.
2. **Empirical structure.** The empirical work falls naturally into three distinct studies, each with its own data, methods, and findings. A standard-format integration would have required compressing or fragmenting the studies in awkward ways.
3. **Career development.** Drafting the constituent papers during the doctoral period accelerates the candidate's transition to academic publishing.

This format choice was made in consultation with the supervisory team during Year 2 of the programme and recorded on the Notice of Submission form.

7.2 Overarching research question

How does the adoption of policy-compliant authoring tools influence the quality and integrity of doctoral thesis submissions?

The question is decomposed into three sub-questions, each addressed by one of the papers:

- **Paper 1** (Chapter 2): *What patterns of adoption do we observe across UK business schools?*
- **Paper 2** (Chapter 3): *Does compliance enforcement causally improve writing quality?*
- **Paper 3** (Chapter 4): *How do supervisors perceive and interpret the role of tooling in supervision?*

7.3 Coherence across the papers

Although each paper stands alone, the three together form a coherent body of work. Paper 1 establishes the empirical landscape of tool adoption; Paper 2 isolates the causal effect of one specific intervention (compliance enforcement); Paper 3 interprets the quantitative findings through the lived experience of supervisors. The Conclusion (Chapter 5) synthesises across the three.

7.4 Statement on the candidate's contribution

The candidate is the first author of all three papers. A detailed contribution declaration is provided at the head of each paper chapter using the `::: {.contribution}` Div, as required by policy section 13.3.

Chapter 8

Paper 1: Adoption Patterns of Compliant Authoring Tools in UK Business Schools

Contribution statement

I designed the study, drafted and piloted the survey instrument, recruited the participating institutions, collected and cleaned the data, performed the analysis, and wrote the manuscript. Co-author A advised on sampling strategy and reviewed successive drafts.

8.1 Abstract

Cross-sectional survey of doctoral programme directors at 28 UK business schools investigating patterns of adoption of policy-compliant authoring tools. We find substantial heterogeneity across institutions, with adoption rates ranging from 8% to 71%.

8.2 Introduction

Policy-compliant authoring tools have proliferated rapidly over the past five years, but their uptake in doctoral education is poorly documented (Smith and Johnson, 2024).

8.3 Methods

A web-administered survey was sent to doctoral programme directors at all 28 AACSB-accredited UK business schools. The response rate was 89% ($n = 25$ usable responses).

8.4 Results

Adoption rates ranged from 8% to 71% with a mean of 34%. Two institutional factors are associated with higher adoption:

- **Institutional size:** a one-standard-deviation increase in cohort size is associated with a 6.2 percentage-point increase in adoption ($p < 0.05$).
- **Dedicated methods training:** programmes with a formal research-methods sequence show 11.4 percentage-point higher adoption ($p < 0.01$).

8.5 Discussion

The findings suggest that adoption is structurally enabled by institutional resources rather than driven by individual candidate preferences.

8.6 Limitations

The cross-sectional design precludes causal claims.

8.7 Appendix 2.A: Survey instrument

The full 18-item survey instrument is reproduced below. Items were piloted with three programme directors and refined based on cognitive interviews.

i Section 1: Programme demographics

Items 1.1–1.4 capture institution size, accreditation status, and doctoral cohort size.

8.8 Appendix 2.B: Sampling frame

The sampling frame was constructed from the AACSB UK directory accessed in September 2025.

Chapter 9

Paper 2: The Effect of Compliance

Enforcement on Writing Quality: An A/B Experiment

Contribution statement

I designed the experimental protocol, secured ethical approval, recruited participants, randomised treatment assignment, collected outcome data, performed the statistical analysis, and wrote the manuscript. Co-author A advised on the experimental design and Co-author B served as a second blind rater.

9.1 Abstract

We report a pre-registered randomised experiment in which 120 doctoral candidates were assigned to write a 5,000-word chapter using either a compliance-enforcing authoring tool (treatment) or an unconstrained alternative (control). Compliance enforcement caused a 0.31 standard-deviation improvement in the blinded quality rating (95% CI: 0.09–0.52, $p = 0.006$).

9.2 Introduction

Paper 1 established that adoption of compliant authoring tools varies systematically across institutions. This paper asks whether adoption *causes* improvements in writing quality through a randomised experiment.

9.3 Methods

9.3.1 Participants and randomisation

Candidates ($n = 120$) were recruited from three participating universities and randomised at the individual level. Randomisation was stratified by programme year and institution.

9.3.2 Outcome measure

The primary outcome is a blinded expert rating of overall writing quality on a 1–7 scale, averaged across two independent raters. Inter-rater reliability was high ($\kappa = 0.78$).

9.4 Results

Treatment caused a statistically significant improvement in the quality rating. Mean quality in the treatment group was 4.92 (SD 0.91); mean in control was 4.61 (SD 0.86).

9.5 Discussion

The experimental design provides causal identification of the effect of compliance enforcement on writing quality.

9.6 Limitations

The 5,000-word chapter task may not generalise to the full thesis-writing process.

9.7 Appendix 3.A: Pre-registration

The full pre-registration of hypotheses, analysis plan, and stopping rule is archived at the Open Science Framework.

9.8 Appendix 3.B: Power analysis

The minimum detectable effect was calculated at 0.25 standard deviations at 80% power, $\alpha = 0.05$.

9.9 Appendix 3.C: Robustness checks

The headline result holds across three pre-registered robustness specifications.

Chapter 10

Paper 3: Supervisor Perspectives on Authoring Tooling in Doctoral Supervision: A Qualitative Study

Contribution statement

I designed the interview protocol, recruited and interviewed all 15 supervisor participants, transcribed the interviews, conducted the primary thematic analysis, and wrote the manuscript. Co-author C served as a second independent coder for inter-rater reliability and reviewed the manuscript.

10.1 Abstract

Semi-structured interviews with 15 doctoral supervisors at three UK business schools examined how supervisors interpret and respond to candidate use of policy-compliant authoring tools. Three themes emerged: *delegation of compliance work*, *displacement of intellectual conversation*, and *re-distribution of authority over thesis structure*.

10.2 Introduction

Papers 1 and 2 establish that policy-compliant tooling is increasingly adopted and that adoption causally improves writing quality. But the quantitative work cannot illuminate the *meaning* of these tools for the supervisory relationship.

10.3 Methods

10.3.1 Sampling

A purposive sample of 15 supervisors was recruited from three universities, balanced by gender, career stage, and disciplinary sub-field.

10.3.2 Analysis

Transcripts were coded using reflexive thematic analysis.

10.4 Findings

10.4.1 Theme: delegation of compliance work

Supervisors framed compliance tooling as a “delegation” that freed the supervisor-candidate conversation from procedural concerns:

Before, I would spend half of our meeting going over the declaration template and the copyright statement. Now the tool does that. We can actually talk about the argument.

— Supervisor 7, senior professor

10.4.2 Theme: displacement of intellectual conversation

A minority of supervisors expressed concern that the tools could *displace* intellectual conversation.

10.4.3 Theme: re-distribution of authority

The most consistent theme was a perceived shift in authority over thesis structure from the supervisor to the institution (via the tool).

10.5 Discussion

The findings complement the quantitative results in Papers 1 and 2 by showing that the *meaning* of tooling adoption is contested.

10.6 Limitations

The sample is restricted to three universities and to business and management disciplines.

10.7 Appendix 4.A: Interview topic guide

The semi-structured interview topic guide developed across the first four interviews is reproduced below.

1. Tell me about your current doctoral supervisees.
2. Are any of your supervisees using a policy-compliant authoring tool?
3. What has been your experience of working with candidates who use these tools?
4. Have there been moments of difficulty or tension related to the tool?
5. Whose decisions does the tool encode? Do you see this as appropriate?

10.8 Appendix 4.B: Coding framework

The coding framework includes three top-level themes and twelve sub-codes.

Chapter 11

Conclusion

This chapter synthesises across the three papers and offers practical recommendations.

11.1 Synthesis of findings

The three papers together establish a coherent picture:

- **Paper 1** documents substantial cross-institutional variation in adoption and identifies institutional size and methods training as enablers.
- **Paper 2** provides causal evidence that compliance enforcement improves writing quality at the chapter level.
- **Paper 3** illuminates the meanings supervisors attach to tooling and identifies tensions around delegation, displacement, and authority.

A common thread is that tooling is most effective when embedded in an institutional infrastructure that includes supervisor engagement.

11.2 Theoretical contribution

The thesis advances the *adoption-as-infrastructure* perspective by treating policy-compliant authoring tools as components of a broader institutional ecology rather than as discrete artifacts adopted by individuals.

11.3 Practical implications

Three practical recommendations follow:

1. **Institutional:** methods-training programmes should integrate authoring-tool training as a first-class component.
2. **Supervisory:** supervisor development should explicitly address the tensions identified in Paper 3.
3. **Tool design:** compliance tools should expose, rather than hide, the structural decisions they enforce.

11.4 Limitations across the papers

The three papers share a UK business-school context. Replication in other national systems and disciplinary cultures is a priority for future work.

11.5 Future research agenda

- A multi-country comparative study.
- A longitudinal panel that tracks candidates from registration to submission.
- A design experiment that tests alternative tool affordances.

11.6 Closing remarks

The findings reinforce an argument from The University of Manchester (2026): that the *infrastructure* of academic writing shapes the integrity of research outputs.

- [] Smith, J.A. and Johnson, M.B. (2024) 'Innovation in postgraduate research', *Journal of Doctoral Education*, 18(3), pp. 245–267. Available at: <https://doi.org/10.1234/jde.2024.18.3>.
- [] The University of Manchester (2026) 'Presentation of Theses Policy, version 12'. Available at: <https://documents.manchester.ac.uk/display.aspx?DocID=7420>.